

● *Protons and Heavy Ions—II*

CLINICAL EXPERIENCE AND EXPECTATION WITH PROTONS AND HEAVY IONS†

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The experience in fractionated radiation therapy using proton beams from the 160 MV Harvard cyclotron on 23 patients is described. Advantages for the proton beam are expected to lie entirely with the dose distribution patterns which are qualitatively different from those obtainable with photon beam techniques. Treatment is discussed in terms of complexity of compensation for tissue heterogeneity and patients are placed into three categories. I: no heterodense structure within the beam path; II: one or two large and heterodense structures of simple configuration; III: multiple and/or complex density heterogeneities within the beam path. Treatment results seem almost certain to be improved for selected sites which would be in Treatment Category I. We are confident that improved results can be obtained for certain of the Category II problems. Additional work must be completed before much in the way of a sensible prediction can be made for patients whose treatment problems would be in Category III. The largest experience to date is the treatment of choroidal melanoma. Because of the high precision of dose delivery and uniformity of dose distribution achieved we are confident that after an appropriate dose level has been selected that good results will be obtained on a regular basis for this group of tumors. General problems of particle therapy are discussed and details of treatment of several illustrative cases are presented.

Fractionated radiation therapy, Particle radiation therapy, Proton beams, Cyclotron, Dose distribution, Photon beams, Tissue compensation, Choroidal melanoma.

INTRODUCTION

High expectations for proton radiation therapy are based exclusively on the physical fact that depth dose characteristics for modulated energy proton beams are qualitatively different from those obtained using photons. The difference in depth dose curves between protons and photons is a consequence of the fact that protons have a finite range in tissue while the depth dose for photons decreases approximately exponentially with depth in tissue. Importantly, the proton range in tissue is energy dependent. By employing appropriate techniques for energy modulation, depth dose curves similar to the one shown in Fig. 1 can be

generated. In the same figure, a depth dose curve for a 42 MV photon beam is included. The proton depth dose curve as shown is flat throughout a desired range in tissue. This range of uniform dose is determined by the patterns of energy modulation. The depth of penetration is, of course, set by the maximum proton energy. A special feature of this depth dose curve is the extraordinarily rapid decrease in dose at the distal end of the beam range. The dose decreases from 80% of peak value to 20% over a range of about 4 mm. Further, the dose across the width of the beam can be made flat, and at the beam edge the dose decreases from 80% to 20% in ap-

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